A Toolbox in Sequences and Series

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In these notes, we will search for patterns within sequences and series and emerge with the tools to surgically dissect related problems with efficiency and confidence. For reference, a sequence is defined as the ordered arrangement of numbers and a series is defined as the sum of the numbers within a sequence. For instance, $1, 1, 2, 3, 5, 8, ...$ is a sequence of numbers—in fact, the ubiquitous Fibonacci Sequence—and $1 + 1 + 2 + 3 + 5 + 8$ is one of its series.

1 Arithmetic Sequence

An arithmetic sequence is a sequence of numbers where the difference between any two consecutive or adjacent terms is constant. We call this difference d the common difference of the sequence. The first term of such a sequence is represented as a_1. Accordingly, the nth term of the sequence, a_n, is given by $\boxed{a_1 + (n-1)d}$. Arithmetic sequences include $\{1, 3, 5, 7, ...\}$, $\{2, 4, 6, 8, ...\}$, and $\{3, 6, 9, 12, ...\}$, for instance.

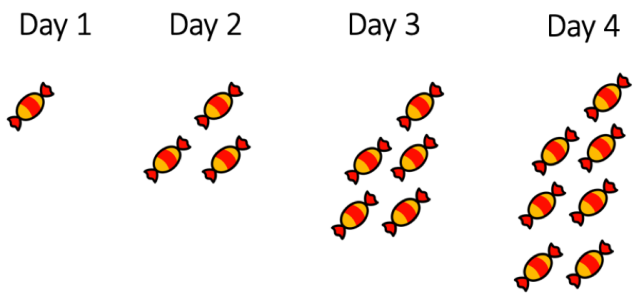


Figure 1: The number of candies as a function of the nth day forms an increasing arithmetic sequence with common difference 2.

Consider the sum of the members of an arithmetic sequence, which we call an arithmetic series—for instance, $2 + 4 + 6 + 8$. We can express the sum S of n consecutive members of a sequence with common difference d and starting with a_1 as the following.

$$S = a_1 + (a_1 + d) + (a_1 + 2d) + ... + (a_1 + (n-2)d) + (a_1 + (n-1)d)$$

Note that $a_1 + (n-1)$ is the nth element of the sequence, so we are essentially taking the sum of all terms between the $1st$ and nth terms, inclusive. If we invert the right hand side, we extract the following.

$$S = (a_1 + (n-1)d) + (a_1 + (n-2)d) + ... + (a_1 + 2d) + (a_1 + d) + a_1$$

We can add the two formulas for S together, matching term a_1 in the first equation with $a_1 + (n-1)d$ in the second equation, and so forth.

$$S + S = \underbrace{[a_1 + (a_1 + (n-1)d)] + [(a_1 + d) + (a_1 + (n-2)d)] + ... + [(a_1 + (n-1)d) + a_1]}_{n \text{ terms each enclosed with square brackets } [\,]}$$

We note that each bracketed term is equal to $2a_1 + (n-1)d$ and we have n of these terms present on the right hand side of the above equation. Thus,

$$2S = n * (2a_1 + (n-1)d)$$
$$S = \frac{n}{2}(2a_1 + (n-1)d).$$

Let $a_n = a_1 + (n-1)d$ be the nth term of the sequence. Then, the sum S can be rewritten as

$$S = \frac{n}{2}(a_1 + a_1 + (n-1)d) = \frac{n}{2}(a_1 + a_n).$$

Example 1

Suppose that $\{a_n\}$ is an arithmetic sequence with

$$a_1 + a_2 + \cdots + a_{100} = 100 \text{ and } a_{101} + a_{102} + \cdots + a_{200} = 200.$$

What is the value of $a_2 - a_1$? (AMC 10B 2002)

Solution: We can subtract both of the provided equations to produce the following.

$$\begin{aligned}(a_{101} + a_{102} + \cdots + a_{200}) - (a_1 + a_2 + \cdots + a_{100}) &= 200 - 100\\ (a_{101} - a_1) + (a_{102} - a_2) + \cdots + (a_{200} - a_{100}) &= 100.\end{aligned}$$

We note that each of the differences enclosed in parentheses—which are of the form $a_{i+100} - a_i$ for $1 \le i \le 100$—is equal to $100d$, where d is the common difference between terms. We note that there are 100 of these expressions.

$$\begin{aligned}100d + 100d + \cdots + 100d &= 100.\\ 100d * 100 &= 100.\end{aligned}$$

Thus, $a_2 - a_1 = d = \frac{1}{100}$, and our final answer is $\boxed{\frac{1}{100}}$.

Example 2

The first four terms in an arithmetic sequence are $x+y$, $x-y$, xy, and $\frac{x}{y}$, in that order. What is the fifth term? (AMC 10B 2003)

Solution: We note that the difference between each of the consecutive terms must be equal. The common difference is $d = a_2 - a_1 = (x-y) - (x+y) = -2y$. Thus, we can express the third term as $a_2 + d = (x-y) + (-2y) = x - 3y$. Similarly, the fourth term is equal to $x - 5y$. The problem states that the third term is xy, so we know that $x - 3y = xy$ so $x - xy = 3y$. Factoring out the x on the left hand side produces $x(1-y) = 3y$ and $x = \frac{3y}{1-y}$.

Analagously, since the fourth term is $\frac{x}{y}$, we find that $\frac{x}{y} = x - 5y$. As $x = \frac{3y}{1-y}$ from an earlier result, then:

$$\begin{aligned}\frac{x}{y} &= x - 5y\\ \frac{\frac{3y}{1-y}}{y} &= \frac{3y}{1-y} - 5y\\ \frac{3}{1-y} = \frac{3y - 5y(1-y)}{1-y} &= \frac{5y^2 - 2y}{1-y}\end{aligned}$$

Multiplying both sides by $1-y$ produces $3 = 5y^2 - 2y$ and thus $5y^2 - 2y - 3 = 0$. This can be factored into $(5y+3)(y-1) = 0$, which enforces that either $y = -\frac{3}{5}$ or $y = 1$. We note that $y = 1$ is impossible since the first four terms of the arithmetic sequence would have to be $x+1$, $x-1$, x, and x in that order, which is contradictory. Thus, $y = -\frac{3}{5}$, which means that $x = \frac{3y}{1-y} = -\frac{9}{8}$. Since

the fourth term is $a_4 = x - 5y$, then the fifth term must be $a_4 + d = (x - 5y) - 2y = x - 7y$, which is equal to

$$x - 7y = -\frac{9}{8} - (7 * -\frac{3}{5}) = -\frac{9}{8} + \frac{21}{5} = \frac{123}{40}.$$

Thus, the answer is equal to $\boxed{\frac{123}{40}}$.

Geometric Sequences

Geometric sequences is similar to arithmetic sequences in that terms follow a rigid pattern. However, instead of a fixed difference between consecutive terms, there is a fixed quotient between them. In other words, given a geometric sequence $\{a_n\}$ and two adjacent terms a_i and a_{i+1}, the value of $\frac{a_{i+1}}{a_i}$ is constant regardless of the choice of $i \geq 1$. An example of a geometric sequence is $\{2, 4, 8, 16, 32, ...\}$, where each element is double the previous element starting with 2.

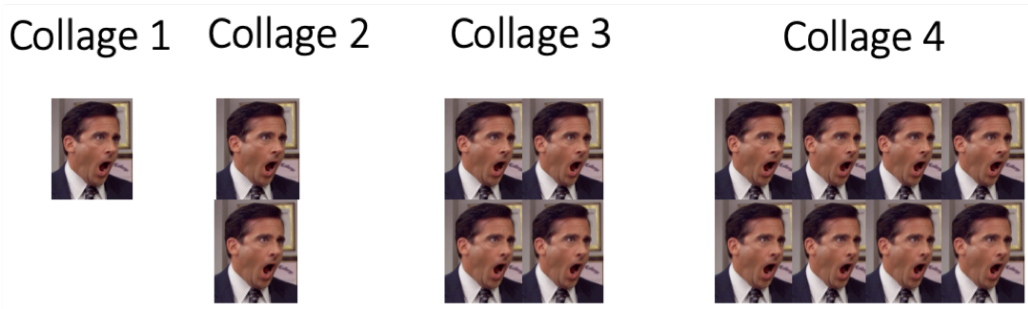


Figure 2: The number of photos in the nth collage represents an increasing geometric sequence with the quantity doubling between adjacent collages.

We call the quotient of consecutive terms, $\frac{a_{i+1}}{a_i}$, the common ratio r of a geometric sequence. Given a starting element a_1, a geometric sequence can be written as $\{a_1, a_1r, a_1r^2, a_1r^3, ...\}$. More succinctly, the nth element can be computed as $\boxed{a_1r^{n-1}}$.

Next, we contemplate the sum of a geometric series with n terms starting with a_1 and common ratio r. This can be written as $S = a_1 + a_1r + a_1r^2 + ... + a_1r^{n-1}$. To find a closed formula, we multiply both sides of the equation by r to produce $Sr = a_1r + a_1r^2 + ... + a_1r^n$. Noting that most of the terms cancel out, subtracting the first equation from the second equation leads to

$$\begin{aligned} Sr - S = (a_1r + a_1r^2 + ... + a_1r^n) - (a_1 + a_1r + a_1r^2 + ... + a_1r^{n-1}) \\ Sr - S = a_1r^n - a_1 \\ S(r-1) = a_1(r^n - 1) \\ S = \frac{a(1-r^n)}{1-r} \end{aligned}$$

.

By principle, the sum of a geometric series starting from first term a_1 with n terms and a common ratio of r is equal to

$$S = \frac{a_1(1-r^n)}{1-r}.$$

Some geometric series are infinite, meaning that they never end just like the Minnesota Viking's playoff misery. Some infinite series do not converge such as $1 + 2 + 4 + 8 + ...$, where the sum expands without bound. However, an infinite series like $1 + \frac{1}{2} + \frac{1}{4} + ...$, where each term is half the preceding term, converges to 2, as shown below.

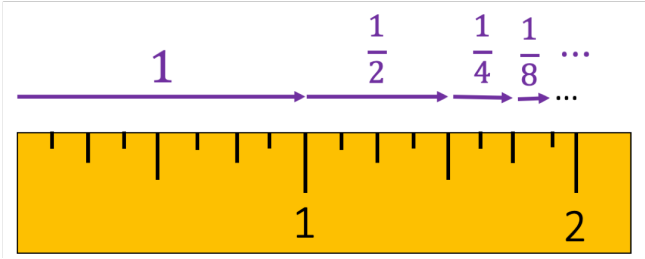


Figure 3: Convergence of the series $1 + \frac{1}{2} + \frac{1}{4} + ...$ to the value 2.

To formalize this arc of understanding, an **infinite geometric series** will converge as long as $|r| < 1$, which means that each succeeding term is closer to 0 than the previous term. Consider the sum presented earlier with $S = \frac{a_1(1-r^n)}{1-r}$. As $n \to \infty$, then r^n will approach zero since $|r| < 1$. While calculus is desired to rigorously prove the result below, we intuitively find that for an infinite series with $n \to \infty$, then

$$S = \frac{a_1(1-r^n)}{1-r} = \frac{a_1(1-0)}{1-r} = \frac{a_1}{1-r}.$$

This is the sum for an infinite geometric series starting with term a_1 with common ratio r. The formula is important result with numerous ramifications for problem-solving strategy, as we will unveil below.

Example 3

The second and fourth terms of a geometric sequence are 2 and 6. Which of the following is a possible first term? (AMC 10B 2003)
(A) $-\sqrt{3}$ **(B)** $-\frac{2\sqrt{3}}{3}$ **(C)** $-\frac{\sqrt{3}}{3}$ **(D)** $\sqrt{3}$ **(E)** 3

Solution: Let a be the first term and r be the common ratio of the sequence. Then, the second term is ar and the fourth term is ar^3, so $ar = 2$ and $ar^3 = 6$. Dividing the two equations produces $\frac{ar^3}{ar} = \frac{6}{2}$, or $r^2 = 3$. Thus, $r = \pm\sqrt{3}$. Since $ar = 2$, then $a = \frac{2}{r} = \frac{2}{\pm\sqrt{3}} = \pm\frac{2\sqrt{3}}{3}$. Thus, the correct answer is $\boxed{\text{B: } -\frac{2\sqrt{3}}{3}}$.

Example 4

For $-1 < r < 1$, let $S(r)$ denote the sum of the geometric series

$$12 + 12r + 12r^2 + 12r^3 + \cdots.$$

Let a between -1 and 1 satisfy $S(a)S(-a) = 2016$. Find $S(a) + S(-a)$. (AIME I 2016)

Solution: The central expression $12 + 12r + 12r^2 + 12r^3 + \cdots$ is an infinite geometric series with starting element 12 and common ratio r, meaning that its sum is equal to $S(r) = \frac{a_1}{1-r} = \frac{12}{1-r}$. Thus,

$$\begin{aligned} S(a)S(-a) &= 2016 \\ \frac{12}{1-a} * \frac{12}{1+a} &= 2016 \\ \frac{144}{1-a^2} &= 2016. \end{aligned}$$

We note that $S(a)+S(-a)=\frac{12}{1-a}+\frac{12}{1+a}=\frac{12(1+a)}{(1-a)(1+a)}+\frac{12(1-a)}{(1+a)(1-a)}=\frac{24}{1-a^2}$. We know from above that $\frac{144}{1-a^2}=2016$, and dividing both sides by 6 yields $\frac{24}{1-a^2}=\frac{2016}{6}=\boxed{336}$, which is precisely the result that we are searching for.

Example 5

Let $a+ar_1+ar_1^2+ar_1^3+\cdots$ and $a+ar_2+ar_2^2+ar_2^3+\cdots$ be two different infinite geometric series of positive numbers with the same first term. The sum of the first series is r_1, and the sum of the second series is r_2. What is r_1+r_2? (AMC 12A 2009)

Solution: The sum of a geometric series with first term a and common ratio r is $\frac{a}{1-r}$. According to this formula, the sum of the two sequences in the problem are $\frac{a}{1-r_1}$ and $\frac{a}{1-r^2}$. Since these are equal to r_1 and r_2, respectively, then we have

$$\frac{a}{1-r_1}=r_1 \quad \text{and} \quad \frac{a}{1-r_2}=r_2$$
$$r_1(1-r_1)=a \quad \text{and} \quad r_2(1-r_2)=a$$

Since $r_1 \neq r_2$, then r_1 and r_2 must be the distinct roots of the equation $x(1-x)=a$. Expanding this out leads to $x^2-x+a=0$, and on the basis of Vieta's Formulas, the sum of the roots is $r_1+r_2=-\frac{-1}{1}=\boxed{1}$.

Example 6

In an increasing sequence of four positive integers, the first three terms form an arithmetic progression, the last three terms form a geometric progression, and the first and fourth terms differ by 30. Find the sum of the four terms. (AIME I 2003)

Solution: Let the first term of this sequence be a and the common differnece of the arithmetic progression be d. Thus, the first three terms can be written as $a, a+d, a+2d$. Since the last three terms constitute a geometric progression with a common ratio of $\frac{a+2d}{a+d}$, the fourth term can be written as $(a+2d)r=(a+2d)*\frac{a+2d}{a+d}=\frac{(a+2d)^2}{a+d}$. Since the first and fourth terms differ by 30 and the sequence is increasing, then $\frac{(a+2d)^2}{a+d}-a=30$. Multiplying both sides by $a+d$ leads to $(a+2d)^2-a(a+d)=30a+30d$, which crystallizes into $3ad+4d^2=30a+30d$ upon simplification.

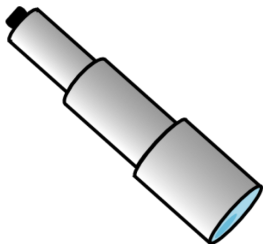
We can rewrite this equation as $4d^2-30d=30a-3ad$, or $2d(2d-15)=3a(10-d)$. Since d and a are both positive, then $2d-15$ and $10-d$ must be either both positive or both negative for the equation to hold true. If $2d-15<0$ and $10-d<0$, then $d<\frac{15}{2}$ and $d>10$, which is a contradiction. Thus, both are positive, meaning that $2d-15>0$ and $10-d>0$, so $d>\frac{15}{2}=7.5$ and $d<10$. Since d is an integer, it is either equal to 8 or 9. Furthermore, from the equation $2d(2d-15)=3a(10-d)$, we note that either $2d$ or $2d-15$ must be divisible by 3. In either case, d must be divisible by 3. Thus, $d=9$ and accordingly, $3a(1)=2*9(2*9-15)=54$, so $a=18$.

The four terms of the sequence are therefore 18, 27, 36, and 48, which sum to $\boxed{129}$.

Telescoping Series

A popular toy or instrument was a collapsible telescope, one that could be fully expanded to peer into the night sky or quickly dilated into a shorter cylinder for convenient storage.

A telescoping series in mathematics is one that can be readily contracted into a shorter form. This is process that typically involves the cancellation of several terms. To understand how to interpret and solve these series, we will start with a canonical example.



Example 7

The following can be represented as a simplified fraction $\frac{m}{n}$, where m and n are positive integers such that $\gcd(m,n) = 1$. Compute $m + n$.

$$\frac{1}{1*2} + \frac{1}{2*3} + \frac{1}{3*4} + ... + \frac{1}{99*100}$$

Solution: We note that each term in the sum is of the form $\frac{1}{n(n+1)}$ for an integer $1 \leq n \leq 99$. Note that $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. This means that the expression can be reconfigured as

$$(\frac{1}{1} - \cancel{\frac{1}{2}}) + (\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}) + (\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}}) + ... + (\cancel{\frac{1}{99}} - \frac{1}{100}) = \frac{1}{1} - \frac{1}{100} = \frac{99}{100}.$$

We acknowledge that several neighboring terms cancel each other out, allowing the series to contract like a collapsing *telescope*. As the sum is $\frac{99}{100}$, then $m = 99$ and $n = 100$, so $m + n = \boxed{199}$. An auxiliary note is that the series $\frac{1}{1*2} + \frac{1}{2*3} + \frac{1}{3*4} + ... + \frac{1}{n*(n+1)} = 1 - \frac{1}{n+1}$ converges to 1 as $n \to \infty$.

Next, we will work with a more arcane telescoping problem.

Example 8

Let $f(x) = x^2(1-x)^2$. What is the value of the sum

$$f\left(\frac{1}{2019}\right) - f\left(\frac{2}{2019}\right) + f\left(\frac{3}{2019}\right) - f\left(\frac{4}{2019}\right) + \cdots + f\left(\frac{2017}{2019}\right) - f\left(\frac{2018}{2019}\right)?$$

Solution: We note that $f(1-x) = (1-x)^2(1-(1-x))^2 = (1-x)^2x^2 = f(x)$. The summative expression above can be rewritten as

$$[f(\frac{1}{2019}) - f(\frac{2018}{2019})] - [f(\frac{2}{2019}) - f(\frac{2017}{2019})] + [f(\frac{3}{2019}) - f(\frac{2016}{2019})] + \cdots + [f(\frac{1009}{2019}) - f(\frac{1010}{2019})]$$

We note that each term inside a pair of square brackets [ ] is of the form $f(a) - f(1-a)$ for some fraction a. Since $f(a) = f(1-a)$ as proved above, then $f(a) - f(1-a) = 0$. Therefore, all of the terms in the sum are equal to 0, meaning that the entire sum is equal to $\boxed{0}$.

Exercises

1. How many terms are in the arithmetic sequence $13, 16, 19, ..., 70, 73$. (AMC 10A 2015)
2. The first three terms of a geometric progression are $\sqrt{3}$, $\sqrt[3]{3}$, and $\sqrt[6]{3}$. What is the fourth term? (AMC 12A 2014)
3. Five positive consecutive integers starting with a have average b. What is the average of 5 consecutive integers that start with b? (AMC 12A 2014)

4. A grocer makes a display of cans in which the top row has one can an each lower row has two more cans than the row above it. If the display contains 100 cans, how many rows does it contain? (AMC 10B 2004)

5. For a design, Margo creates n concentric circles out of yarn. The first and smallest circle has a radius of 2 cm. Each circle after that has a radius that is 1 cm greater than the previous circle. For gluing purposes she needs 0.5 cm of extra yarn for each circle. If $f(n)$ represents the total amount of yarn needed for n circles, express $f(n)$ in terms of $f(n-1)$ and n. (MN Math League 15/16)

6. Determine exactly the value of

$$\frac{1}{\sqrt{2}-\sqrt{1}}+\frac{1}{\sqrt{3}-\sqrt{2}}+\cdots+\frac{1}{\sqrt{400}-\sqrt{399}}$$

(MN Math League 15/16)

7. Let $a < b < c$ be three integers such that a, b, c is an arithmetic progression and a, c, b is a geometric progression. What is the smallest possible value of c? (AMC 12A 2014)

8. Find

$$\frac{2^2}{2^2-1} * \frac{3^2}{3^2-1} * \frac{4^2}{4^2-1} \cdots \frac{2006^2}{2006^2-1}.$$

(HMMT 2006)

9. Jo and Blair take turns counting from 1 to one more than the last number said by the other person. Jo starts by saying "1", so Blair follows by saying "$1, 2$" . Jo then says "$1, 2, 3$" , and so on. What is the 53rd number said? (AMC 10B 2013)

10. A sequence of three real numbers forms an arithmetic progression with a first term of 9. If 2 is added to the second term and 20 is added to the third term, the three resulting numbers form a geometric progression. What is the smallest possible value for the third term in the geometric progression? (AMC 12A 2004)

11. One hundred concentric circles with radii $1, 2, 3, \ldots, 100$ are drawn in a plane. The interior of the circle of radius 1 is colored red, and each region bounded by consecutive circles is colored either red or green, with no two adjacent regions the same color. The ratio of the total area of the green regions to the area of the circle of radius 100 can be expressed as m/n, where m and n are relatively prime positive integers. Find $m + n$. (AIME I 2003)

12. How many non-similar triangles have angles whose degree measures are distinct positive integers in arithmetic progression? (AMC 10A 2006)

13. Two geometric sequences $a_1, a_2, a_3, \ldots$ and $b_1, b_2, b_3, \ldots$ have the same common ratio, with $a_1 = 27$, $b_1 = 99$, and $a_{15} = b_{11}$. Find a_9. (AIME II 2012)

14. How many sets of two or more consecutive positive integers have a sum of 15? (AMC 12A 2006)

15. An infinite geometric series has sum 2005. A new series, obtained by squaring each term of the original series, has 10 times the sum of the original series. The common ratio of the original series is $\frac{m}{n}$ where m and n are relatively prime integers. Find $m + n$. (AIME II 2005)

16. Two distinct, real, infinite geometric series each have a sum of 1 and have the same second term. The third term of one of the series is $1/8$, and the second term of both series can be written in the form $\frac{\sqrt{m}-n}{p}$, where m, n, and p are positive integers and m is not divisible by the square of any prime. Find $100m + 10n + p$. (AIME II 2002)

17. Consider the sequence $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, ...$, where the integer n appears n times. Compute the 1992nd term of this sequence. (ARML 1992)

18. Let $a_1, a_2, ...$ be an arithmetic sequence and $b_1, b_2, ...$ be a geometric sequence. Suppose that $a_1b_1 = 20$, $a_2b_2 = 19$, and $a_3b_3 = 14$. Find the greatest possible value of a_4b_4. (HMMT 2019)

19. The sum of an infinite geometric series is a positive number S, and the second term in the series is 1. What is the smallest possible value of S? (AMC 10B 2006)

20. The terms of an arithmetic sequence add to 715. The first term of the sequence is increased by 1, the second term is increased by 3, the third term is increased by 5, and in general, the kth term is increased by the kth odd positive integer. The terms of the new sequence add to 836. Find the sum of the first, last, and middle terms of the original sequence. (AIME I 2012)

21. The sequences of positive integers $1, a_2, a_3, ...$ and $1, b_2, b_3, ...$ are an increasing arithmetic sequence and an increasing geometric sequence, respectively. Let $c_n = a_n + b_n$. There is an integer k such that $c_{k-1} = 100$ and $c_{k+1} = 1000$. Find c_k. (AIME II 2016)

22. Let $z = 1 - 2i$. Find $\frac{1}{z} + \frac{2}{z^2} + \frac{3}{z^3} + ...$ (HMMT 2003)

23. The vertices of a quadrilateral lie on the graph of $y = \ln x$, and the x-coordinates of these vertices are consecutive positive integers. The area of the quadrilateral is $\ln \frac{91}{90}$. What is the x-coordinate of the leftmost vertex? (AMC 12A 2020)

24. Let $ABCD$ be a unit square. Let Q_1 be the midpoint of $\overline{CD}$. For $i = 1, 2, \ldots,$ let P_i be the intersection of $\overline{AQ_i}$ and $\overline{BD}$, and let Q_{i+1} be the foot of the perpendicular from P_i to $\overline{CD}$. What is
$$\sum_{i=1}^{\infty} \text{Area of } \triangle DQ_iP_i\,?$$
(AMC 12B 2016)

25. There exists a unique strictly increasing sequence of nonnegative integers $a_1 < a_2 < \ldots < a_k$ such that
$$\frac{2^{289}+1}{2^{17}+1} = 2^{a_1} + 2^{a_2} + \ldots + 2^{a_k}.$$
What is k? (AMC 12A 2020)

26. Show that the cube roots of three distinct prime numbers cannot be three terms (not necessarily consecutive) of an arithmetic progression. (USAMO 1973)

27. Prove that the average of the numbers $n \sin n^\circ$ $(n = 2, 4, 6, \ldots, 180)$ is $\cot 1^\circ$. (USAMO 1996)